KARAN KAILAS MENDHE

Electrical Engineering Undergraduate | Embedded Systems | Power Electronics | IoT & Hardware Prototyping

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PROFESSIONAL SUMMARY

Electrical Engineering undergraduate (B.Tech, expected 2027) with hands-on experience building hardware-software systems end to end — from embedded control on ESP32/Arduino platforms to power electronics circuits and IoT-based monitoring systems. Led development of a wearable robotics device and an IoT renewable-energy monitoring platform, working across component selection, firmware, sensor integration, and hardware testing. National finalist at Smart India Hackathon 2025 and Smartathon 2.0 (IEEE), with hands-on industrial exposure through an internship at Electrical Loco Shed, Nagpur.

TECHNICAL SKILLS

Programming: C, C++, Python

Embedded Systems & Microcontrollers: ESP32, Arduino, Arduino IDE, Sensors & Actuators, Relay & Motor Control, Firmware Development

Electronics & Hardware: Circuit Design, PCB Fundamentals, Hardware Prototyping, Electronic Testing, Soldering, Power Electronics, Electrical Machines, Industrial Automation, Basic Power Systems

IoT & Communication: IoT System Design, MQTT, Cloud Connectivity (Firebase), Mobile App Integration

CAD, PCB & Simulation: AutoCAD Electrical, MATLAB, Simulink, LTspice, Tinkercad, Altium Designer (basic PCB design)

PROJECTS

Dextra — Smart Hand Orthosis Sep 2025 – Mar 2026

Wearable ESP32-controlled hand orthosis for wrist/finger rehabilitation — Smart India Hackathon 2025 Grand Finalist, Team Dexteriest (Medtech/Healthcare, top 5 teams nationally)

- Conceived the project and led a 6-person team building a **460g** wearable orthosis, selecting and validating components — dual servo actuators, tendon-driven mechanism, 4000mAh Li-ion battery — against grip-force (**~56N**) and weight targets
- Developed ESP32 firmware supporting three control methods (three-way switch, mobile app, voice commands), iterating from an initial switch-only build to a more reliable multi-input system, and removing a relay module identified as a reliability bottleneck
- Documented the system design and presented the technical build at the SIH 2025 Grand Finale

REYNEX — Renewable Energy & Microgrid Monitoring System Dec 2025 – Apr 2026

IoT-based ESP32 monitoring platform for hybrid solar/microgrid systems — Smartathon 2.0 (IEEE) National-Level Finalist, Team Raynex (Renewable Energy, top 10 teams nationally)

- Built and programmed ESP32 firmware integrating **5 sensor types** (ACS712 current, INA219 voltage/current, PZEM-004T energy meter, DC voltage, temperature) for real-time monitoring
- Designed an offline-first architecture with cloud sync (Firebase, MQTT) and a companion mobile app for real-time data and alerts
- Tested and calibrated sensor accuracy against a multimeter through repeated calibration passes, on a 5-person team

High Gain DC–DC Boost Converter Aug 2024 – Dec 2025

High-gain, two-input DC–DC boost converter for EV and renewable-energy applications

- Designed and simulated a converter stepping up a low-voltage DC input to a stable **~360V** output at **20kHz** PWM switching
- Fabricated and validated a two-input PCB hardware prototype against simulation results, refining the design to reduce voltage stress and switching losses

INDUSTRIAL TRAINING

Electrical Loco Shed, Nagpur — Industrial Trainee Jan 16 – Jan 22, 2025

- Observed and studied locomotive electrical systems, including traction motors and battery banks, in a live industrial maintenance environment
- Gained exposure to industrial maintenance procedures and electrical safety practices through direct observation of real-world operations

EDUCATION

Bachelor of Technology (B.Tech), Electrical Engineering

Bajaj Institute of Technology, Wardha — Expected Graduation: 2027 | CGPA: 6.60

CERTIFICATIONS

- MATLAB Onramp — MathWorks
- Simulink Onramp — MathWorks
- Introduction to Power Electronics — Coursera (University of Colorado Boulder)
- PCB Basic Design Course — Altium Education
- One-Week Short-Term Training Program (STTP): Smart Grid Control – Renewable Energy & Vehicle Integration

ACHIEVEMENTS

- **Smart India Hackathon 2025** — Grand Finalist, Team Dexteriest (Medtech/Healthcare) — top 5 teams selected nationally
- **Smartathon 2.0** — National-Level Finalist, Team Raynex (Renewable Energy) — top 10 teams selected nationally
- **Wardha Innovation Challenge** — Runner-up — district-level startup competition judged on product and market viability

ADDITIONAL

Relevant Coursework: Power Electronics, Electrical Machines, Control Systems, Power Systems, Industrial Automation, Embedded Systems, Analog & Digital Electronics

Skills: Problem Solving, Analytical Thinking, Teamwork, Leadership, Effective Communication, Time Management, Adaptability, Attention to Detail | **Languages:** English, Hindi, Marathi